Insulin sensitivity in Chinese ovo-lactovegetarians compared with omnivores.

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AIM: To compare the insulin sensitivity indices between Chinese vegetarians and omnivores. METHODS: The study included 36 healthy volunteers (vegetarian, n=19; omnivore, n=17) who had normal fasting plasma glucose levels. Each participant completed an insulin suppression test. We compared steady-state plasma glucose (SSPG), fasting insulin, the homeostasis model assessment for insulin sensitivity (HOMA-IR and HOMA %S) and beta-cell function (HOMA %beta) between the groups. We also tested the correlation of SSPG with years on a vegetarian diet. RESULTS: The omnivore subjects were younger than the vegetarians (55.7+/-3.7 vs 58.6+/-3.6 year of age, P=0.022). There was no difference between the two groups in sex, blood pressure, renal function tests and lipid profiles. The omnivores had higher serum uric acid levels than vegetarians (5.25+/-0.84 vs 4.54+/-0.75 mg/dl, P=0.011). The results of the indices were different between omnivores and vegetarians (SSPG (mean+/-s.d.) 105.4+/-10.2 vs 80.3+/-11.3 mg/dl, P<0.001; fasting insulin, 4.06+/-0.77 vs 3.02+/-1.19 microU/ml, P=0.004; HOMA-IR, 6.75+/-1.31 vs 4.78+/-2.07, P=0.002; HOMA %S, 159.2+/-31.7 vs 264.3+/-171.7%, P=0.018) except insulin secretion index, HOMA %beta (65.6+/-18.0 vs 58.6+/-14.8%, P=0.208). We found a clear linear relation between years on a vegetarian diet and SSPG (r=-0.541, P=0.017). CONCLUSIONS: The vegetarians were more insulin sensitive than the omnivore counterparts. The degree of insulin sensitivity appeared to be correlated with years on a vegetarian diet.